

GL-ProQ-en-006

Incoming Inspection Procedure

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Revision: 3.0

Effective Date: 02/02/2026

Revision Date: 02/02/2026

Purpose

Define key elements of incoming quality inspection with external suppliers on both standard and custom parts, used in the manufacture of products or services.

Updated with a change to inspection requirements based on part criticality, with a move from *detection* to *prevention*.

Recognize capable suppliers (have process maturity & data visibility) with reduced internal SE inspection effort as skip lot status. MIQS suppliers are so excluded.

Frees SE site resources to focus on systemic quality improvement

Scope

- All SE Manufacturing plants (GSC, ETO) & LAC, but not DCs.
- Material received from all external suppliers, used in SE products and services for Customers. Brand labelled products are excluded.

Target Audience

- Supply Quality Engineers
- Quality Engineers
- Goods Receivers
- Incoming inspectors
- Supplier leaders

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References

- ISO:2859-1 - Sampling procedures for inspection by attributes
 - ISO:3951-2 - Sampling procedures for inspection by variables
 - [Supplier Excellence guide](#)
 - GL-DirQ-en-049 Measuring Means Monitoring Directive
 - GL-GuiQ-en-011 Measurement System Analysis (MSA)
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Process Inputs

- Sub-assemblies, products and materials
-

Process Overview

The incoming inspection step is important to secure Schneider Electric manufacturing plants, it is critical for our business to minimize customer impact, as offer safety alerts or claims.

The incoming inspection (refer to [process map in appendix A](#)) includes:

- the expectations from [Inspection area management](#)
- the [content of incoming inspection plan](#) with theoretical data needed as:
 - criticality coming from Line of business / Industrialization & SAP
 - data for good inward requirements coming from supplier quality engineer as sample Inspection & Dynamic rules
- the advice for right selection for data to inspect ([Attributes versus variables](#))
- the [sample size rules](#) (depending on type of data)
- the way to qualify means through [Gage R&R](#)
- the rules for cases of recurrent non-conformance ([Quality Wall](#))

Incoming Inspection must be looked at as an action in response to a risk and consequently:

- the risk, including the risk for the customers, must be evaluated by appropriate means,
- the inspection shall be adapted to the risk level.

As described in Supplier Excellence Guide (especially chapter 7 Part Product qualification), the supplier is responsible for carrying out on-going tests and inspections of its production to ensure continuing conformance to Schneider Electric's requirements. In case of non-conformity due to supplier (refer to Supplier Excellence Guide, especially chapter 8 Supplier Performance monitoring), all costs must be recorded and passed to procurement (non-limited examples: red time for SE, sorting parts, TQW, increased defect rate on production line, extra rework, ...) and take actions (refer to Supplier Excellence Guide, sub-chapter 8-1 Non-conformance management).

The objective of this procedure is to establish the rules, roles, and responsibilities relative to reception, verification, of the parts/material before using it in Schneider-Electric manufacturing lines.

**Inspection Area
Management**

All materials delivered are subject to basic commercial checks to identify them and confirm them against the Purchase Order in the ERP and booked as received. No material may be removed from this area until confirmed.

All material confirmed, in the ERP or other systems, as not requiring Goods Inwards Inspection can be passed to the Warehouse and stored as 'Available Stock', respecting 5S standards, for the safety of our workers and quality of our goods, as well as FIFO.

All other materials must be passed to Goods Inwards Inspection. A random sample will be taken, as defined in this document, from all containers in the delivery to ensure it is representative. The sample will then be inspected against the Specification, Drawing or Control Plan for conformance.

All conforming materials can be passed to the Warehouse and stored as Available Stock.

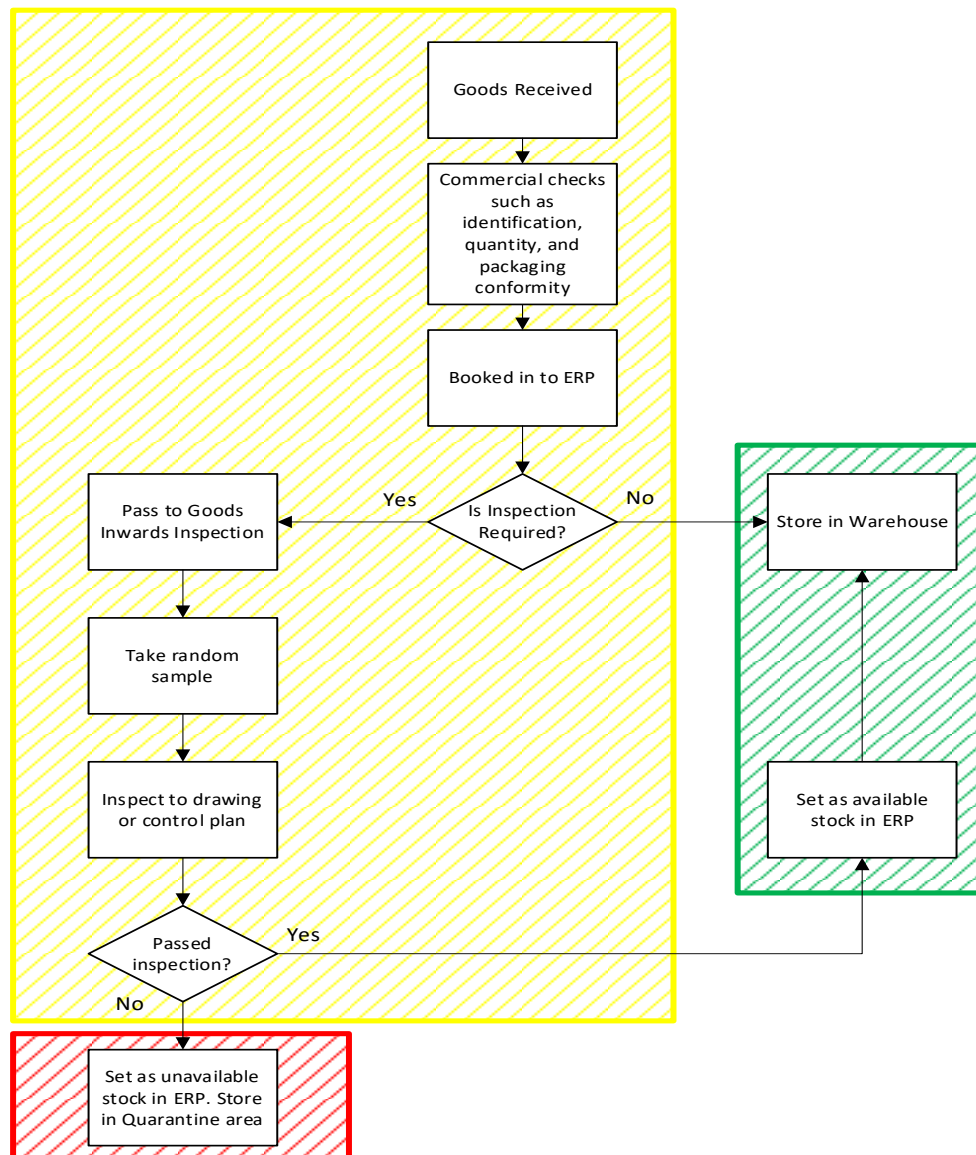
A non-conforming material refers to 2 types of defects : commercial or technical:

- A commercial defect refers to quantity errors, mislabelling, packaging issues, etc ..., which doesn't affect the quality of the part / product. If nonconformity is easily rectified, then Quarantine area is not mandatory
- A technical defect refers to which affect fit, form, function or potentially pass-through features with end effect to customer and/or local effect to Schneider Electric COPQ (high MDR, low FPY, ...)

All non-conforming materials (excepting commercial defects already solved) must be placed in a Quarantine Area, awaiting disposition instructions, such as: Return to Supplier, Derogation, 100% Sorting or Scrap. The Quarantine Area must be only accessible by authorised personnel. No material may be removed from the Quarantine Area with the authority of the Quality Department.

**Inspection Area
Management
(cont.)**

The Goods Receiving and Goods Inwards Inspection should be clearly zoned as shown below, and 5S respected at all times:



**Incoming
inspection plan**

1. Incoming inspection plan must define “What”, “How many”, “How” if it must be inspected.

This should be completed at a characteristics level, if possible. If characteristic criticality is not defined, then the part severity will be used to determine the sample size for all characteristics.

2. Characteristics criticality or part severity and assessment

The line of Business must assess:

The criticality of key characteristics of the part/material, sub-assembly:

- A – Critical Characteristic CTQ S/R (Critical To Quality, Safety & Regulation)
- B – Major Characteristic CTQ (Critical To Quality, Working)
- C – Minor characteristic

The part severity according:

- High (ranking 9-10) Regulation & Property Damage/Safety
- Medium (ranking 4-7) Performance/Working
- Low (ranking 1) Imperceptible

The organisations within Schneider Electric shall respect OLM (Offer Lifecycle Management) rules to assess severity of parts and characteristics in a consistent manner throughout the organization based on the FMEA (Failure Mode and Effects Analysis) principles.

Any legacy parts/characteristics, that do not have a criticality/severity defined, then it can be set by the plant quality team, with support from the Technical Antenna & LOB. The criticality or severity should be based on experience, knowledge, and historical data. In case of any doubt, the criticality/severity must be set to the highest level.

Incoming inspection must focus first on criticality A then B (or High then Medium (if only severity available)) for the customer and manufacturing.

The incoming inspection can be implemented in three ways (depending on history of controls and criticality / severity):

- 100% inspection
- sampling inspection
- no inspection

3. Sample Inspection & Dynamic rules

The probationary period is meant to gain confidence in the supplier's ability to deliver goods with sustained quality and to optimize and validate incoming inspection conditions.

It relies on various supplier manufacturing batches.

Full batch traceability is mandatory to be attached in ERP. If no traceability, then each delivery shall be considered as a new batch.

Rules for direct skip lot (2 options) :

1. **If full APQP with PPAP approval** (all PPAP elements completed , no PPAP deviation, no open action, or no expired validation) then no probationary period is required, skip lot is allowed.

If any characteristic is still pending validation, then apply TQW methodology (with traceability ensured) to ensure product robustness (refer to APQP rules).

For-critical characteristic / part, a certificate of analysis is required for each supplier manufacturing batch (format is under plant / supplier control, it could be digitized or paper), supplier must always be able to deliver data details on SE request (if needed) : no data => no skip lot.

Supplier to maintain **ongoing PPAP validity**; requalification required after process, material, or tooling changes.

2. **When no PPAP available** (mainly legacy parts), if reliable data show supplier performance is defect free during probationary period, then skip lot is allowed.

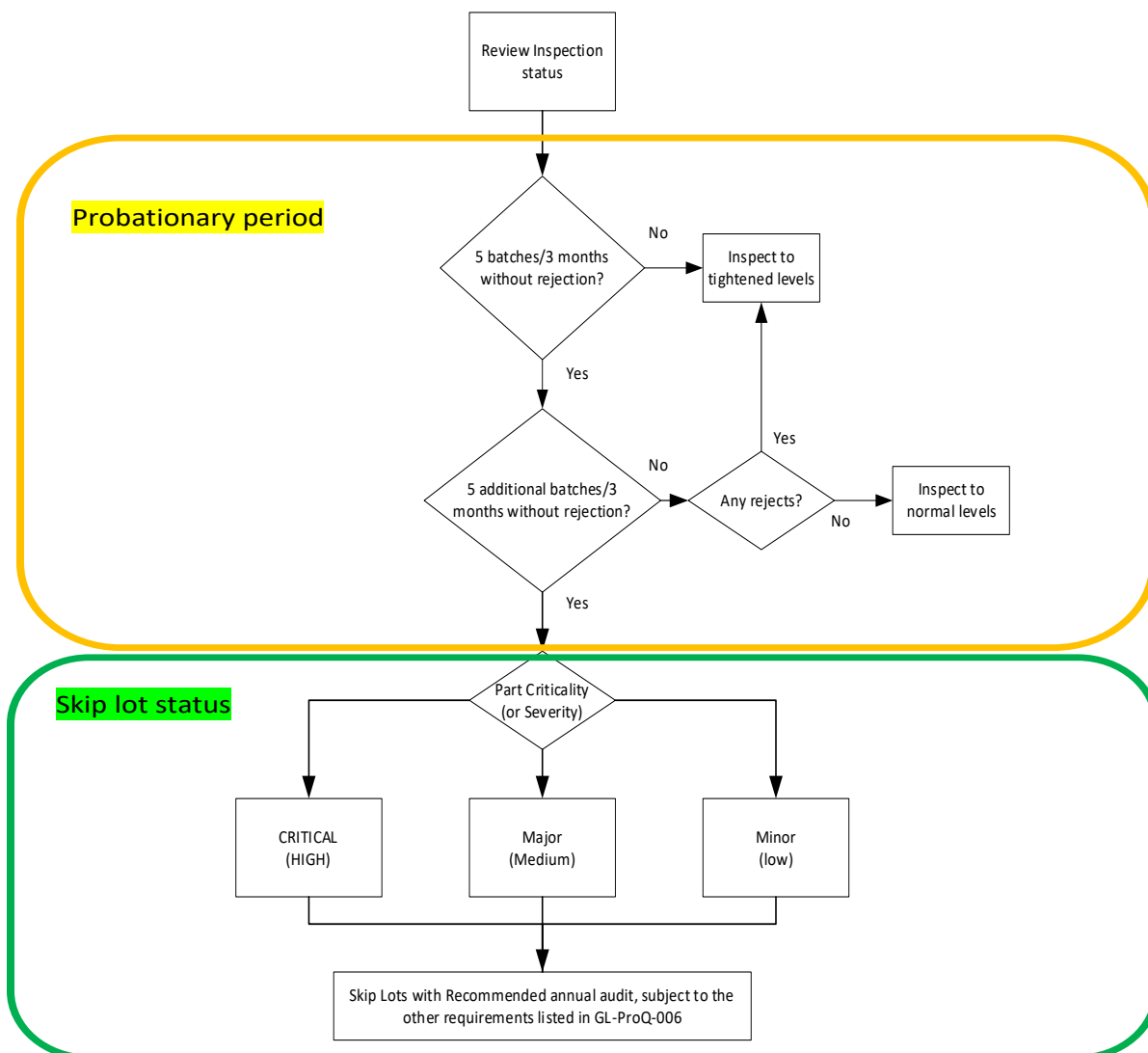
For-critical characteristic / part, a certificate of analysis is required for each supplier manufacturing batch (format is under plant / supplier control, it could be digitized or paper), supplier must always be able to deliver data details on SE request (if needed) : no data => no skip lot.

If part / characteristic is not within one of 2 previous rules, all deliveries are subject to incoming inspection for a defined probationary period.

1. The probationary period lasts up to 3 months and involves at least 5 deliveries (one period **at normal level**).
2. After this normal level period with **only** conforming technical material, then skip lot is allowed.
3. If any rejection during normal level period, then the probationary period starts again **at tightened level**. It will last up to 3 months and involves at least 5 deliveries
4. After this tightened level period with **only** conforming technical material, then skip lot is allowed.

For-critical characteristic / part, a certificate of analysis is required for each supplier manufacturing batch (format is under plant / supplier control, it could be digitized or paper), supplier must always be able to deliver data details on SE request (if needed) : no data => no skip lot.

General



Rules to maintain skip lot :

1. If a technical incident (means, except commercial defect listed above) occurs in SE production and/or found by customer, **skip lot status is immediately suspended.**
2. An appropriate action plan, approved by Schneider Electric, which has been effectively implemented, according I2P (Issue To Prevention), is required
3. A new probationary period of 5 batches / 3 months maximum (**at tightened level!**) will be used, and if **only** conforming technical material inspected during that period, then skip lot is allowed.
4. **For-critical characteristic / part, a certificate of analysis is required for each supplier manufacturing batch (format is under plant / supplier control, it could be digitized or paper), supplier must always be able to deliver data details on SE request (if needed) : no data => no skip lot.**

However, for long Supply Chain procurement (ex: sea transportation), the recommendation (not mandatory) is to maintain permanent inspection.

Attributes
versus variables

As much as possible, inspection by Variables should be preferred to inspection by Attributes.

It must be kept in mind that the difference between Attributes and Variables is dictated by the way the inspection results are managed:

Inspection Means	Part per part decision Conforming / non- conforming	Global Decision on the sample Statistical (average, sigma)
Measurement device (calliper, 3D CMM, amperemeter)	VARIABLES (not recommended)	VARIABLES
Attribute type device (go / no go gauge)	ATTRIBUTES	N/A

Sample Size

1. **For attribute inspection** the ISO:2859-1 standard should be applied for the sample size (it refers to controls such as colour, visual inspection & Go/No Go gage or where each sample is compared only to the limit and not the other samples)

Accept the batch if 0 defect is found in the sample (A=0)
Reject the batch if 1 or more defect is found in the sample (R=1)

Considering the inspection cost, this is the **MINIMUM** sampling plan to follow:

	Critical	Major	Minor
Reduced	20	13	5
Normal	32	20	13
Tightened	50	32	20

Schneider Electric has defined 3 levels of AQL (Acceptance Quality Level) based on criticality of the characteristic (or severity)

- A – HIGH : 0.4%
- B – Medium : 0.65%
- C – low : 1%

2. **For variables inspection (statistical)** the ISO:3951-2 standard should be applied for the sample size. The minimum sample size for all part criticalities is 10 pieces, when using statistical methods (it will give an estimation of Cpk).

- A – HIGH : Cpk ≥ 1.67
- B – Medium : Cpk ≥ 1.33
- C – low : Cpk ≥ 1.0

- If the measurement result, where the Cpk is close to the expected value, then further samples can be inspected to confirm the result.

The sampling quantity should be based on the manufacturing batch size, if known and identified, otherwise it should be based on the delivered batch size. The skip lot rules should also be applied to the manufactured batch, if known and identified.

Gage R&R

Gage R&R (Repeatability & Reproducibility) must be performed on measurement devices, to ensure they are suited to the measurement task. The Gage R&R should be repeated for any significant changes, that may affect measurement accuracy. Refer to GL-GuiQ-en-011 Measurement System Analysis (MSA).

Quality Wall

In cases of recurrent non-conformance, where the supplier clearly does not have sufficient control over its production process, the Quality Wall will be applied as part of the containment. An external company (Third party inspector) validated by Schneider Electric will implement sorting in line on 100% with the criteria defined at the supplier's site. The cost of sorting must be captured and provided to procurement team. Sorting results will be communicated to both Schneider Electric and the supplier. Quality wall should be removed once corrective actions have been proven to be efficient.

Rules for Quality wall inspection:

- 1 - Inspection shall be concentrated on the defects encountered for the last 6 months – called Firewall Defects.
 - 2 - Inspect by measuring, examining, testing or gauging for such defects and comparing the results with the requirements to assess conformity.
 - 3 - Any additional defects seen in the Quality wall inspection shall be added as Firewall defects.
 - 4 - Firewall defects may be graduated if such defect is no longer being encountered for the succeeding consecutive three months
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Records

Records of inspection decisions must be kept for at least 5 years after receipt of materials or goods, unless customer requirements or regulations specify a greater period.

The 5 minimum necessary data are detailed in [Quality Records Section](#).

Other data to be recorded will be based on the needs to provide traceability, as defined by the LOB. It is preferred to maintain records of actual measurement data, where variable data is available.

Process Outputs

Inspection decisions and measurement data that can be used to analyze supplier performance and ensure product quality.

Monitoring & Control

- means monitoring from quality / metrology (refer to GL-DirQ-en-049 Measuring Means Monitoring Directive)
- appropriate KPIs (as % of suppliers / parts on skip-lot, Skip-lot reversion rate, DPPM trend (pre/post, % of suppliers integrated into SPC monitoring, Supplier reaction cycle time, Cost of recovery transferred to suppliers, ...) coming from SQE monitoring with monthly performance analysis of suppliers

Roles & Responsibilities

Role	Responsibilities
Process Owner	Monitor effectiveness of this process Review adequacy of this document
LOB	Define criticality of parts and characteristics
SQE	Define inspection plan requirements (any change) Execute & track skip-lot & update SPC data (monthly)
Incoming Inspector	Inspect parts according to the inspection plans and keep records
Goods Receiver	Pass all deliveries requiring inspection, to the Goods Inwards Inspection department and correctly storing goods after inspection

Competency Requirements

Role	Knowledge	Competency Level	Intervention
Process Owner	Directive Subject Matter	Competent	Training, education & experience
LOB	Tolerancing, DFMEA, Part Criticality, SAP, PDM	Competent	Training, education & experience
Supplier Quality Engineer (SQE)	Quasar IQC, SAP, Tolerancing	Competent	Training and education
Incoming Inspector	Measuring equipment, Quarantine, SAP, Quasar IQC, Tolerancing	Competent	Training
Goods Receiver	SAP, Quarantine rules, Material Handling	Competent	Training

Quality Records

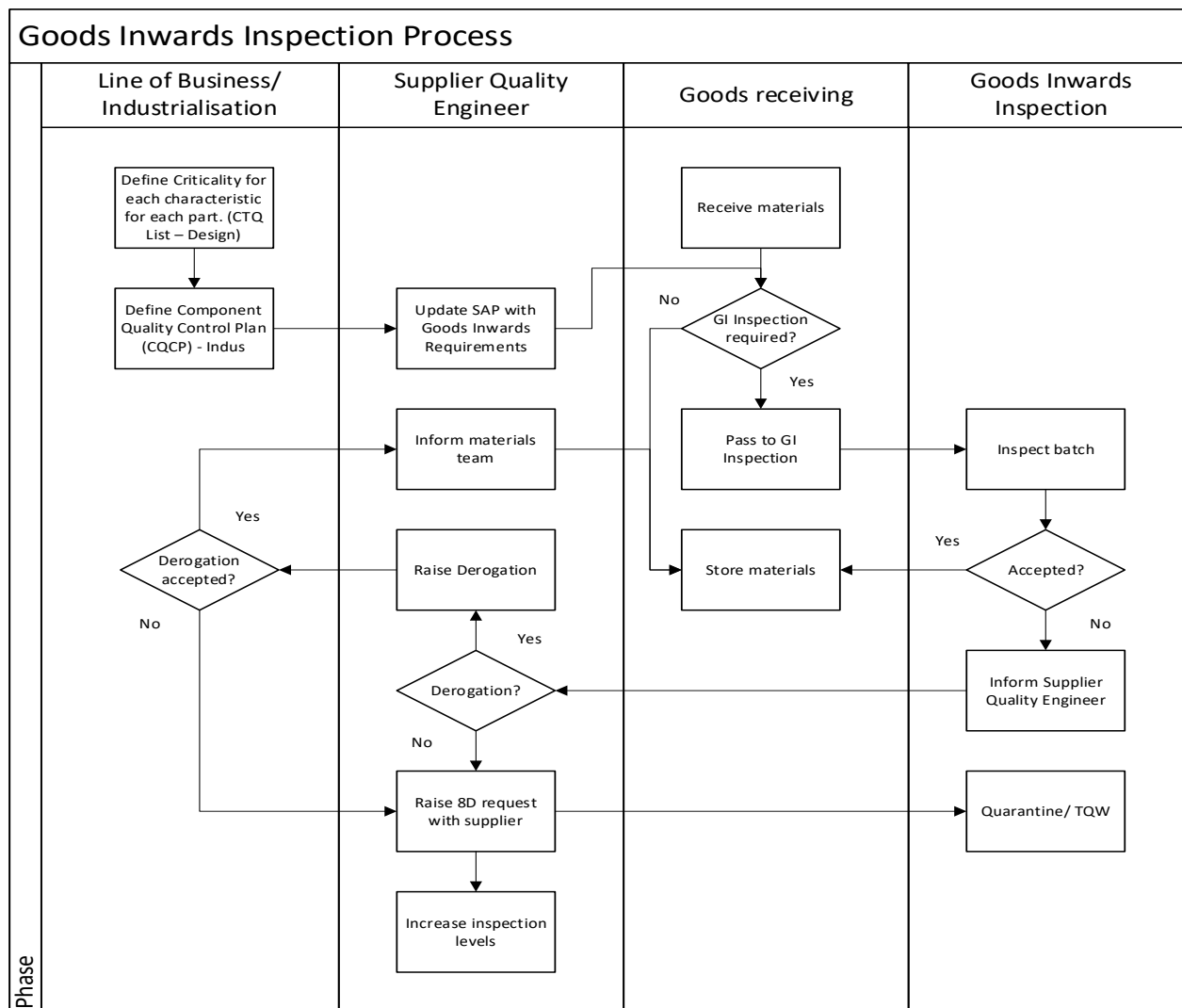
Record	Record Location	Retention Period
Part Number	Plant/Network	5 years minimum
supplier manufacturing batch number		
Inspection Date		
Quantity Received		
Inspection Decision		

**Terms &
Acronyms**

The table below includes definitions for terms and acronyms used in this document. Reference **ISEE** for a full list of terms and acronyms used in the SE IMS.

Term	Definition
BIIS	Balanced Incoming Inspection Sampling
CTQ	Critical To Quality
CTQ S/R	Critical To Quality Safety or Regulation
DC	Distribution Center
DPPM	Defective Parts Per Million
GSC	Global Supply Chain
ETO	Engineering To Order
LOB	Line of Business
LAC	Local Adaptation Center
MIQS	Most Impacting Quality Supplier
SQE	Supplier Quality Engineer
MSA	Measurement Systems Analysis (Gage R&R)
OLM	Offer Lifecycle Management
FMEA	Failure Modes Effects Analysis
CPK	Process Capability Index
I2P	Issue to Prevention

Appendix A: Process Map



General

REVISION HISTORY SUMMARY

Approvers	Requested By	Change Description	Reason for Change	Revision	Revision Date	Contributors
ERIC JACQUOT; NEVILLE DUCIE	BERTRAND COURQUIN	<u>1. Purpose & Philosophy</u> • Old: Focused on detection through incoming inspection • New: Shift from detection to prevention, recognizing capable suppliers and enabling skip-lot status for those with process maturity and data visibility <u>2. Skip-Lot & Dynamic Rules</u> • Old: Probationary period (3–6 months, 5 deliveries) without full skip-lot criteria • New: Detailed skip-lot rules: ◦ Direct skip-lot allowed if full APQP with PPAP approval ◦ Mandatory certificate of analysis (COA) for critical parts ◦ Clear reversion rules if incidents occur <u>3. Supplier Accountability</u> • Old: Supplier responsible for non-conformance costs • New: Explicit requirement to record and transfer costs to procurement; integrated with Supplier Excellence Guide <u>4. Inspection Area Management</u>	<u>Estimated Advantages</u> • <u>Quality Improvement: Performance based approach</u> : ◦ focus on prevention and systemic issues rather than detection only. ◦ Improves audit readiness. ◦ Enables proactive performance management • <u>Traceability & Compliance:</u> Enhanced data requirements and ERP integration • <u>Supplier Development:</u> Incentivizes process maturity and continuous improvement. Accelerates supplier qualification,	3.0	02/02/2026	NEIL HOLLIS

		• Old: Quarantine for all non-conforming materials • New: Differentiates commercial vs. technical defects; quarantine mandatory only for technical defects <u>5. Data & Traceability</u> • Old: Basic record keeping (supplier batch number, part number, date, quantity). • New: Adds mandatory batch traceability in ERP, and preference for variable data <u>6. KPIs & Monitoring</u> • Old: Generic KPIs (PPM, DP1) • New: Proposal of advanced KPIs: ◦ % of suppliers/parts on skip-lot, Skip-lot reversion rate, DPPM trend (pre/post), Cost recovery transferred to suppliers, Supplier reaction cycle time <u>7. Roles & Responsibilities</u> • Old: Defined roles (Process Owner, SQE, Inspector) • New: Adds SQE responsibility for skip-lot execution and SPC data tracking				
ERIC JACQUOT	BERTRAND COURQUIN	General updates Clarifications for incoming inspection rules	Obsolescence with mistakes in referentials Missing incoming inspection management	2.0	06/18/2025	

		Adding incoming inspection management				
This request is automatically approved.	BERTRAND COURQUIN	* This document has been migrated from the Global QMS Documents Platform (SharePoint) without revision changes. Process owner/Document author and revision details are specified in the document itself. * A summary of previous revisions is available only in the migrated version.	Document migration	1.0	07/03/2013	